About the Data

This dataset is from NOAA (found here). It contains information about significant earthquakes. The data contains 6,273 unique rows of earthquakes. Each record contains:

- **id:** A unique number assigned to each earthquake.
- **flag_tsunami:** If a tsunami was recorded.
- **year:** Century and year of the significant earthquake. Format +/yyyy (is B.C, +is A.D.). Valid values: 2000 to present. The Date and Time are given in Universal Coordinated Time.
- **month:** Month of the significant earthquake. Valid values: 112. The Date and Time are given in Universal Coordinated Time. The local date may be one day different.
- **day:** Day of the significant earthquake. Valid values: 131 (where months apply). The Date and Time are given in Universal Coordinated Time. The local date may be one day different.
- **hour:** Hour of the significant earthquake. Valid values: 023. The Date and Time are given in Universal Coordinated Time. The local date may be one day different.
- **minute:** Minute of the significant earthquake. Valid values: 059. The Date and Time are given in Universal Coordinated Time. The local date may be one day different.
- **second:** Second of the significant earthquake. Valid values: 059. The Date and Time are given in Universal Coordinated Time. The local date may be one day different.
- **focal_depth:** The depth of the earthquake is given in kilometers. Valid values 0 to 700 km.
- **eq_primary:** The primary earthquake magnitude is chosen from the available magnitude scales in this order: Mw Magnitude, Ms Magnitude, Mb Magnitude, Ml Magnitude, Mfa Magnitude, Unknown Magnitude
- **eq_mag_mw:** Earthquake magnitude Mw. Valid values 0 to 9.9 The Mw magnitude is based on the moment magnitude scale.
- **eq_mag_ms:** Earthquake magnitude MS. Valid values 0 to 9.9 The Ms magnitude is the surfacewave magnitude of the earthquake.
- **eq_mag_mb:** Earthquake magnitude mb. Valid values 0 to 9.9 The Mb magnitude is the compressional body wave (Pwave) magnitude.
- **eq_mag_ml:** Earthquake magnitude ML. Valid values 0 to 9.9 The ML magnitude was the original magnitude relationship defined by Richter and Gutenberg for local earthquakes in 1935.
- **eq_mag_mfa:** Earthquake magnitude Mfa (based on felt area). Valid values 0 to 9.9 The Mfa magnitudes are computed from the felt area, for earthquakes that occurred before seismic instruments were in general use.
- **eq_mag_unk**: Earthquake magnitude type is Unknown. Valid values 0 to 9.9 The computational method for the earthquake magnitude was unknown and could not be determined from the published sources.
- **intensity:** The effect of an earthquake on the Earth's surface is called the intensity. The Modified Mercalli Intensity (MMI) is given in Roman Numerals (converted to numbers in the digital database)
- **country:** Country. The name of the country where earthquake was located.
- **state:** The State, Province or Prefecture of the earthquake was located.

- **location_name:** Earthquake Location Name. The location (city, state or island) where the earthquake was located.
- **latitude:** Latitude: Valid values: 90 to +90 Latitude: 0 to 90 (Northern Hemisphere), 90 to 0 (Southern Hemisphere). The latitude of the location (city, state or island) where the earthquake was located.
- **longitude:** Longitude. Valid values: 180 to +180 Longitude: 0 to 180 (Eastern Hemisphere), 180 to 0 (Western Hemisphere). The longitude of the location (city, state or island) where the earthquake was located.
- **region_code:** Region Name (Code). Regional boundaries were assigned based on the frequency of occurrence of earthquakes, geophysical relations, risk in distant areas and political justification. The codes are defined as: 10 = Central, Western and S. Africa 15 = Northern Africa 20 = Antarctica 30 = East Asia 40 = Central Asia and Caucasus 50 = Kamchatka and Kuril Islands 60 = S. and SE. Asia and Indian Ocean 70 = Atlantic Ocean 80 = Bering Sea 90 = Caribbean 100 = Central America 110 = Eastern Europe 120 = Northern and Western Europe 130 = Southern Europe 140 = Middle East 150 = North America and Hawaii 160 = South America 170 = Central and South Pacific.
- **deaths:** Number of deaths from the earthquake, it may also include deaths caused by the secondary effects such as the tsunami, volcanic eruption or landslide that was triggered by the earthquake.
- **deaths_description:** Description of Deaths from the earthquake. Valid values: 0 to 4. When a description was found in the historical literature instead of an actual number of deaths, this value was coded and listed in the Deaths_amount_order column. If the actual number of deaths was listed, a descriptor was also added for search purposes according to the following definition. 0 = None 1 = Few (~1 to 50 deaths) 2 = Some (~51 to 100 deaths) 3 = Many (~101 to 1000 deaths) 4 = Very Many (~1001 or more deaths)
- **missing:** Number of missing from the earthquake, it may also include missing caused by the secondary effects such as the tsunami, volcanic eruption or landslide that was triggered by the earthquake.
- **missing_description:** Description of Deaths from the earthquake. Valid values: 0 to 4. When a description was found in the historical literature instead of an actual number of missing, this value was coded and listed in the Missing_amount_order column. If the actual number of missing was listed, a descriptor was also added for search purposes according to the following definition. 0 = None 1 = Few (~1 to 50 missing) 2 = Some (~51 to 100 missing) 3 = Many (~101 to 1000 missing) 4 = Very Many (~1001 or more missing)
- **injuries:** Number of injuries from the earthquake, it may also include deaths caused by a secondary effect such as the tsunami, volcanic eruption or landslide that was triggered by the earthquake.
- **injuries_description:** Description of injuries from the earthquake. Valid values: 0 to 4. When a description was found in the historical literature instead of an actual number of injuries, this value was coded and listed in the Injuries_amount_order column. If the actual number of injuries was listed, a descriptor was also added for search purposes according to the following definition. 0 = None 1 = Few (~1 to 50 injuries) 2 = Some (~51 to 100 injuries) 3 = Many (~101 to 1000 injuries) 4 = Very Many (~1001 or more injuries)

- **damage_millions_dollars:** Damage in Millions of Dollars from the earthquake. The damage amount may also include damage caused by a secondary effect such as a tsunami, volcanic eruption, or landslide that was triggered by the earthquake. The value in the Damage column should be multiplied by 1,000,000 to obtain the actual dollar amount in U.S. dollars. The dollar value listed is the value at the time of the event.
- **damage_description:** Description of Damage from the earthquake. For those events not offering a monetary evaluation of damage, the following fivelevel scale was used to classify damage and was listed in the Damage column. If the actual dollar amount of damage was listed, a descriptor was also added for search purposes. 0 = NONE 1 = LIMITED (roughly corresponding to less than $1 million) 2 = MODERATE (~$1 to $5 million) 3 = SEVERE (~>$5 to $24 million) 4 = EXTREME (~$25 million or more) When possible, a rough estimate was made of the dollar amount of damage based upon the description provided, in order to choose the damage category. In many cases, only a single descriptive term was available. These terms were converted to the damage categories based upon the author's apparent use of the term elsewhere.
- **houses_destroyed:** Number of Houses Destroyed. Whenever possible, number of houses destroyed by the earthquake are listed; it may also include houses destroyed caused by a secondary effect such as the tsunami, volcanic eruption, or landslide that was triggered by the earthquake.
- **houses_destroyed_description:** Description of Houses Destroyed by the Earthquake. For those earthquakes not offering an exact number of houses destroyed, the following fourlevel scale was used to classify the destruction and was listed in the Houses_amount_order column. If the actual number of houses destroyed was listed, a descriptor was also added for search purposes. 0 = None 1 = Few (~1 to 50 houses) 2 = Some (~51 to 100 houses) 3 = Many (~101 to 1000 houses) 4 = Very Many (~1001 or more houses)
- **houses_damaged:** Number of Houses Damaged. Whenever possible, number of houses damaged by the earthquake are listed; it may also include houses damaged caused by a secondary effect such as the tsunami, volcanic eruption, or landslide that was triggered by the earthquake.
- **houses_damaged_description:** Description of Houses Damaged by the Earthquake. For those earthquakes not offering an exact number of houses damaged, the following fourlevel scale was used to classify the damage and was listed in the Houses_damaged_amount_order column. If the actual number of houses destroyed was listed, a descriptor was also added for search purposes. 0 = None 1 = Few (~1 to 50 houses) 2 = Some (~51 to 100 houses) 3 = Many (~101 to 1000 houses) 4 = Very Many (~1001 or more houses)
- **total_deaths:** Number of deaths from the earthquake and secondary effects such as the tsunami, volcanic eruption or landslide.
- **total_deaths_description:** Description of Deaths from the earthquake and the secondary effects such as the tsunami, volcanic eruption or landslide. Valid values: 0 to 4. When a description was found in the historical literature instead of an actual number of deaths, this value was coded and listed in the Deaths_amount_order_total column. If the actual number of deaths was listed, a descriptor was also added for search purposes

according to the following definition. 0 = None 1 = Few (~1 to 50 deaths) 2 = Some (~51 to 100 deaths) 3 = Many (~101 to 1000 deaths) 4 = Very Many (~1001 or more deaths)

- **total_missing:** Number of missing from the earthquake and secondary effects such as the tsunami, volcanic eruption or landslide.
- **total_missing_description:** Description of Deaths from the earthquake and secondary effects such as the tsunami, volcanic eruption or landslide. Valid values: 0 to 4. When a description was found in the historical literature instead of an actual number of missing, this value was coded and listed in the Missing_amount_order_total column. If the actual number of missing was listed, a descriptor was also added for search purposes according to the following definition. 0 = None 1 = Few (~1 to 50 missing) 2 = Some (~51 to 100 missing) 3 = Many (~101 to 1000 missing) 4 = Very Many (~1001 or more missing)
- **total_injuries:** Number of injuries from the earthquake and secondary effects such as the tsunami, volcanic eruption or landslide.
- **total_injuries_description:** Description of injuries from the earthquake and secondary effects such as the tsunami, volcanic eruption or landslide. Valid values: 0 to 4. When a description was found in the historical literature instead of an actual number of injuries, this value was coded and listed in the Injuries_amount_order_total column. If the actual number of injuries was listed, a descriptor was also added for search purposes according to the following definition. 0 = None 1 = Few (~1 to 50 injuries) 2 = Some (~51 to 100 injuries) 3 = Many (~101 to 1000 injuries) 4 = Very Many (~1001 or more injuries)
- **total_damage_millions_dollars:** Damage in Millions of Dollars from the earthquake and secondary effects such as the tsunami, volcanic eruption or landslide. The value in the Damage column should be multipled by 1,000,000 to obtain the actual dollar amount in U.S. dollars. The dollar value listed is the value at the time of the event.
- **total_damage_description:** For those events not offering a monetary evaluation of damage, the following fivelevel scale was used to classify damage and was listed in the Damage column. If the actual dollar amount of damage was listed, a descriptor was also added for search purposes. 0 = NONE 1 = LIMITED (roughly corresponding to less than $1 million) 2 = MODERATE (~$1 to $5 million) 3 = SEVERE (~>$5 to $24 million) 4 = EXTREME (~$25 million or more) When possible, a rough estimate was made of the dollar amount of damage based upon the description provided, in order to choose the damage category. In many cases, only a single descriptive term was available. These terms were converted to the damage categories based upon the author’s apparent use of the term elsewhere.
- **total_houses_destroyed:** Number of Houses Destroyed by the earthquake and secondary effects such as the tsunami, volcanic eruption or landslide.
- **total_houses_destroyed_description:** Description of Houses Destroyed by the earthquake and secondary effects such as the tsunami, volcanic eruption or landslide. For those earthquakes not offering an exact number of houses destroyed, the following fourlevel scale was used to classify the destruction and was listed in the Houses_amount_order_total column. If the actual number of houses destroyed was listed, a descriptor was also added for search purposes. 0 = None 1 = Few (~1 to 50

houses) 2 = Some (~51 to 100 houses) 3 = Many (~101 to 1000 houses) 4 = Very Many (~1001 or more houses)

- **total_houses_damaged:** Number of Houses Damaged by the earthquake and secondary effects such as the tsunami, volcanic eruption or landslide.
- **total_houses_damaged_description:** Description of Houses Damaged by the earthquake and secondary effects such as the tsunami, volcanic eruption or landslide. For those earthquakes not offering an exact number of houses damaged, the following fourlevel scale was used to classify the damage and was listed in the Houses_dam_amount_order_total column. If the actual number of houses destroyed was listed, a descriptor was also added for search purposes. 0 = None 1 = Few (~1 to 50 houses) 2 = Some (~51 to 100 houses) 3 = Many (~101 to 1000 houses) 4 = Very Many (~1001 or more houses)

Problem Statement: Significant earthquakes are not evenly distributed around the world. Understanding where earthquakes are likely to cause the most damage helps governments, researchers, and emergency responders be prepared for disasters in high-risk areas.

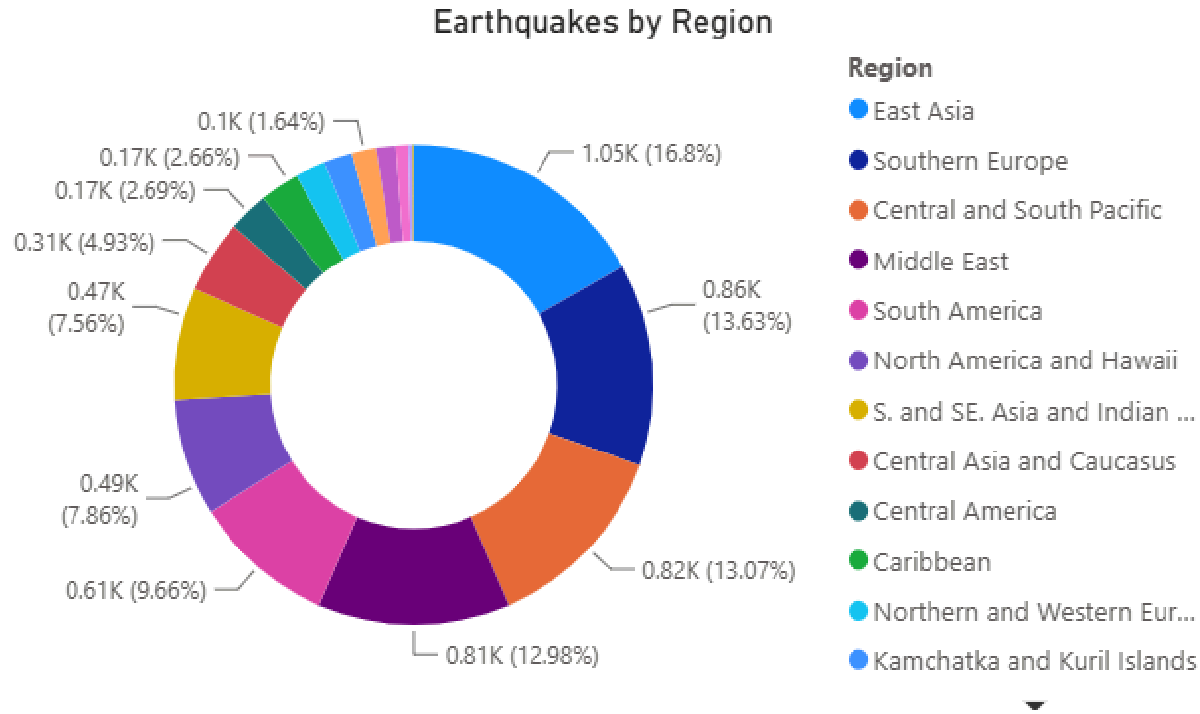
Business Questions

- Which regions experience the highest number of significant earthquakes?
- How has the frequency of significant earthquakes changed over time?
- Where are the global hotspots for significant earthquakes?
- Which regions experience the highest number of tsunami-generating earthquakes compared to non-tsunami earthquakes?
- Which individual earthquakes caused the highest number of deaths, and what were their year, location, intensity, and tsunami occurrence?

Data Cleaning

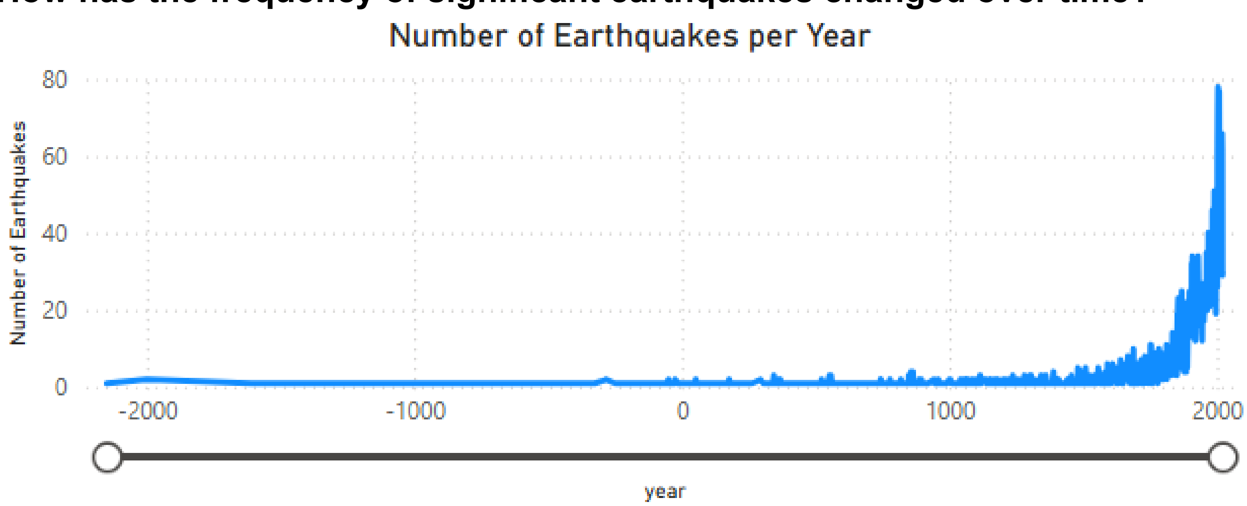
I began the data cleaning by dropping duplicates and checking for missing data. There were no duplicates, but a lot of missing data. Some columns were not necessary for my analysis, so I dropped them. I dropped all magnitude columns except the combined eq_primary column. I also created a column that converted the region_code to region_name that is more useful for analysis. I then multiplied the columns containing the damages in millions of dollars by one million to reflect the actual values.

Which regions experience the highest number of significant earthquakes?



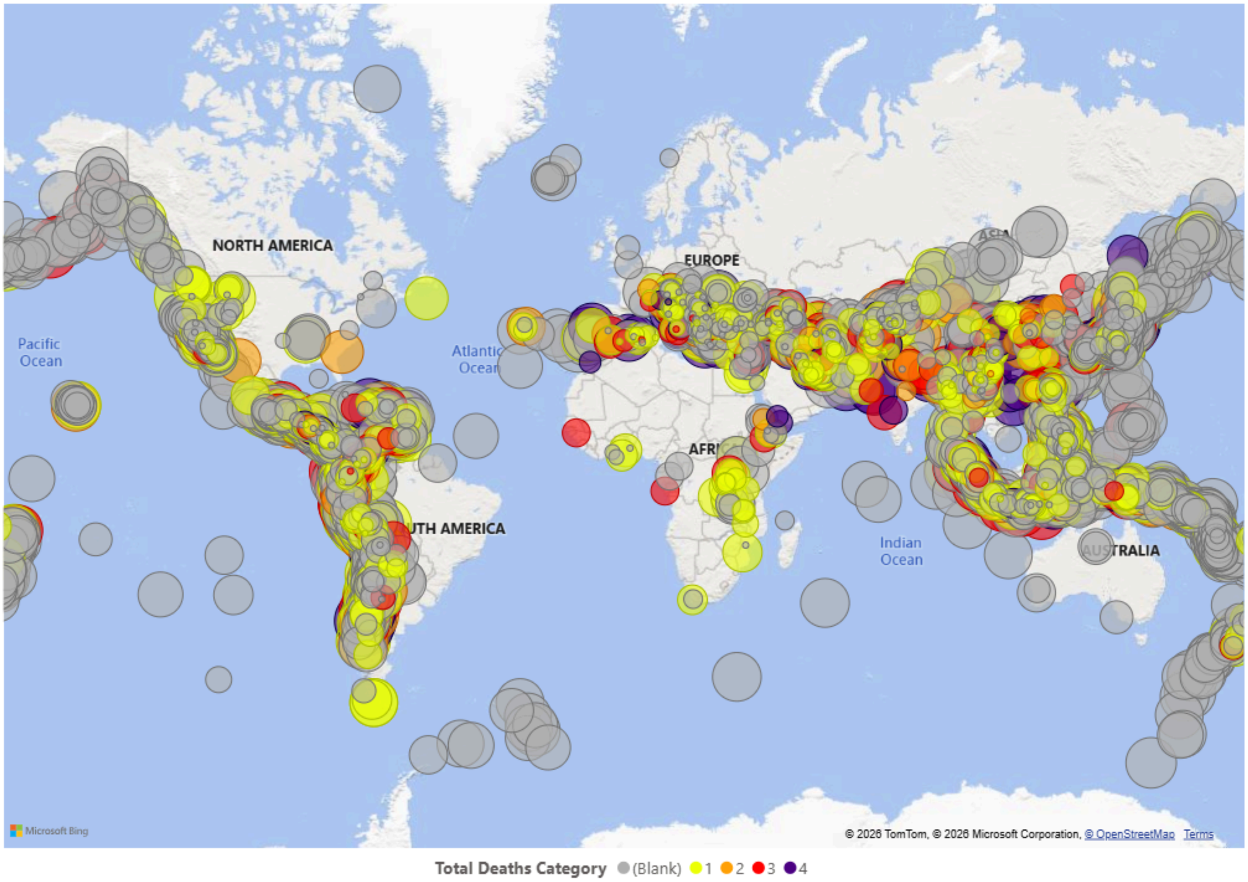
This chart shows that over half of the significant earthquakes occurred in 4 regions (East Asia, Southern Europe, Central and South Pacific, Middle East). Those 4 regions are all located near intersections of tectonic plates. They also generally have a high population, which results in more impactful earthquakes.

How has the frequency of significant earthquakes changed over time?



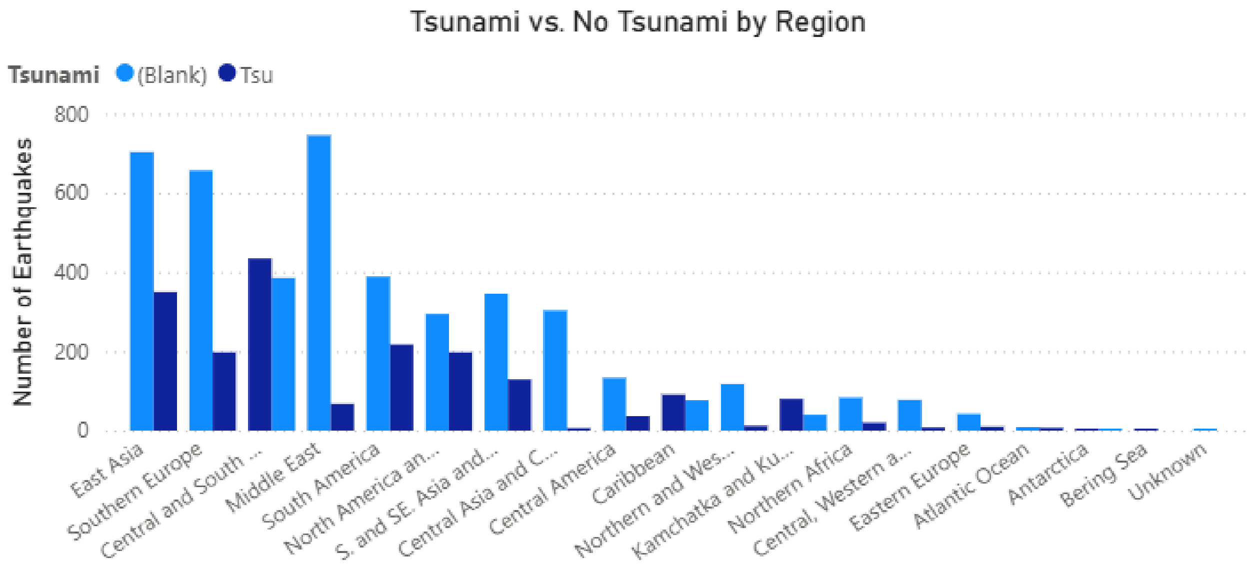
This chart shows that the frequency of significant earthquakes has increased over time. This is largely due to better record keeping starting in the late 1800s. It could also be attributed to population increase resulting in more devastating earthquakes.

Where are the global hotspots for significant earthquakes?



This map shows earthquakes sized by magnitude and colored by total deaths. As expected, most are located along major faultlines and population centers. Those near population centers have higher death counts.

Which regions experience the highest number of tsunami-generating earthquakes compared to non-tsunami earthquakes?



This chart shows that most regions have more non-tsunami than tsunami earthquakes. Most ocean-based regions (Central and South Pacific, Caribbean, Kamchatka and Kuril Islands) have a higher proportion of tsunami-generating earthquakes.

Which individual earthquakes caused the highest number of deaths, and what were their year, location, intensity, and tsunami occurrence?

Location	Year	Intensity	Tsunami	Total Deaths
CHINA: SHAANXI PROVINCE	1556	11		830000
HAITI: PORT-AU-PRINCE	2010		Tsu	316000
TURKEY: ANTAKYA (ANTIOCH)	115	11	Tsu	260000
TURKEY: ANTAKYA (ANTIOCH), SAMANDAG	525	9		250000
CHINA: NE: TANGSHAN	1976	11		242769
AZERBAIJAN: GYZNDZHA	1139	11		230000
IRAN: DAMGHAN, QUMIS	856	10		200000
CHINA: GANSU PROVINCE, SHANXI PROVINCE	1920	12	Tsu	200000
TURKEY: ANTAKYA (ANTIOCH)	458	9		80000
Total				**3368769**

This table shows the earthquakes that caused the highest number of deaths. They occurred in a wide spread of time. All of the deadliest earthquakes have an intensity of 9 or above. They are all located in Asia or the Middle East.

Conclusion and Recommendations

The purpose of this project was to analyze the NOAA Significant Earthquakes dataset to better understand the global distribution and impact of significant earthquakes. This project explored where earthquakes are likely to occur and cause significant damage. These insights can be used to improve disaster preparedness, emergency response, and future research.

The analysis identified where significant earthquakes occur most frequently and visualized where impactful seismic activity is concentrated. The project also compared the occurrence of tsunami-generating earthquakes with non-tsunami earthquakes and identified the individual earthquakes that resulted in the greatest loss of life, as well as key characteristics such as location, time, and intensity.

In this project, I applied the complete data analysis process, including data cleaning, EDA, visualization, and interpretation. Working with a large, real dataset allowed me to understand the importance of data quality and analysis in producing meaningful results. It also strengthened my technical skills in data preparation, SQL querying, and visualization while demonstrating how data can be used to solve real-world problems.

This project supports future analysis to better understand significant earthquakes. Future work could incorporate real-time data or machine learning to explore relationships and patterns between earthquakes and their impacts. This project shows how data analysis can drive insights and solutions to real-world problems and transform records into actionable insights that support decisions.